

Task 1: Git Reset — Hands-On

Make 3 commits in your practice repo (commit A, B, C)

```
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git log --oneline
571091a (HEAD -> master) Modified login_page.html
54a6f57 Modified ai_integration.py
c9c5456 Modified login.html
0d38823 hotfix2
633b5cf added login.html
3a85323 Added file8.html
5a14da8 Added file7.html
d1e4082 Squashed commit of the following:
0d02246 Added file4.py
28844d7 Merge branch 'feature-signup'
da4bdb5 Added file6.html
5c4d7a0 (feature-signup) Added file2.html
5d764c8 Added file.html
8388cc5 Added python script ai_integration.py
9dbe926 (feature-login) Adding healthcheck script
a3ffa5c Added Login Page html file
484c603 Added code for login feature
539809b Update hello-world.sh
3ce969d Added index.html file
c3d31f2 Added script greet.sh
c4ba176 My First Shell script
```

```
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git reset --soft 54a6f57
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git log --oneline
54a6f57 (HEAD -> master) Modified ai_integration.py
c9c5456 Modified login.html
0d38823 hotfix2
633b5cf added login.html
3a85323 Added file8.html
5a14da8 Added file7.html
d1e4082 Squashed commit of the following:
0d02246 Added file4.py
28844d7 Merge branch 'feature-signup'
da4bdb5 Added file6.html
5c4d7a0 (feature-signup) Added file2.html
5d764c8 Added file.html
8388cc5 Added python script ai_integration.py
9dbe926 (feature-login) Adding healthcheck script
a3ffa5c Added Login Page html file
484c603 Added code for login feature
539809b Update hello-world.sh
3ce969d Added index.html file
c3d31f2 Added script greet.sh
c4ba176 My First Shell script
ubuntu@ip-172-31-30-155:~/devops-git-practice$ |
```

```
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   login_page.html

ubuntu@ip-172-31-30-155:~/devops-git-practice$
```

Use git reset --soft to go back one commit — what happens to the changes?

When git reset --soft was used , commits are reverted changes are **kept in the Staging Area**.

```
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git commit -m "Modified login_page.html"
[master 8377b66] Modified login_page.html
1 file changed, 1 insertion(+)
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git log --oneline
8377b66 (HEAD -> master) Modified login_page.html
54a6f57 Modified ai_integration.py
c9c5456 Modified login.html
0d38823 hotfix2
633b5cf added login.html
3a85323 Added file8.html
5a14da8 Added file7.html
d1e4082 Squashed commit of the following:
0d02246 Added file4.py
28844d7 Merge branch 'feature-signup'
da4bdb5 Added file6.html
5c4d7a0 (feature-signup) Added file2.html
5d764c8 Added file.html
8388cc5 Added python script ai_integration.py
9dbe926 (feature-login) Adding healthcheck script
a3ffa5c Added Login Page html file
484c603 Added code for login feature
539809b Update hello-world.sh
3ce969d Added index.html file
c3d31f2 Added script greet.sh
c4ba176 My First shell script
```

git reset --mixed

```
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git reset --mixed 54a6f57
Unstaged changes after reset:
M   login_page.html
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git log --oneline
54a6f57 (HEAD -> master) Modified ai_integration.py
c9c5456 Modified login.html
0d38823 hotfix2
633b5cf added login.html
3a85323 Added file8.html
5a14da8 Added file7.html
d1e4082 Squashed commit of the following:
0d02246 Added file4.py
28844d7 Merge branch 'feature-signup'
da4bdb5 Added file6.html
5c4d7a0 (feature-signup) Added file2.html
5d764c8 Added file.html
8388cc5 Added python script ai_integration.py
9dbe926 (feature-login) Adding healthcheck script
a3ffa5c Added Login Page html file
484c603 Added code for login feature
539809b Update hello-world.sh
3ce969d Added index.html file
c3d31f2 Added script greet.sh
c4ba176 My First shell script
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git status
on branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   login_page.html

no changes added to commit (use "git add" and/or "git commit -a")
ubuntu@ip-172-31-30-155:~/devops-git-practice$ |
```

Re-commit, then use `git reset --mixed` to go back one commit — what happens now?

Following observations are seen :

Commit is reverted

Staging area is cleared , files changes are untracked

Conclusion:

This moves the changes out of the Staging Area and back into your **Working Directory**. The files are still there, but Git marks them as "modified" and "untracked."

```
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git add login_page.html
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git commit -m "Modified login_page.html"
[master d4fda11] Modified login_page.html
1 file changed, 1 insertion(+)
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git log --oneline
d4fda11 (HEAD -> master) Modified login_page.html
54a6f57 Modified ai_integration.py
c9c5456 Modified login.html
0d38823 hotfix2
633b5cf added login.html
3a85323 Added file8.html
5a14da8 Added file7.html
d1e4082 Squashed commit of the following:
0d02246 Added file4.py
28844d7 Merge branch 'feature-signup'
da4bdb5 Added file6.html
5c4d7a0 (feature-signup) Added file2.html
5d764c8 Added file.html
8388cc5 Added python script ai_integration.py
9dbe926 (feature-login) Adding healthcheck script
a3ffa5c Added Login Page html file
484c603 Added code for login feature
539809b Update hello-world.sh
3ce969d Added index.html file
c3d31f2 Added script greet.sh
c4ba176 My First Shell Script
ubuntu@ip-172-31-30-155:~/devops-git-practice$ |
```

```
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git reset --hard 54a6f57
HEAD is now at 54a6f57 Modified ai_integration.py
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git log --oneline
54a6f57 (HEAD -> master) Modified ai_integration.py
c9c5456 Modified login.html
0d38823 hotfix2
633b5cf added login.html
3a85323 Added file8.html
5a14da8 Added file7.html
d1e4082 Squashed commit of the following:
0d02246 Added file4.py
28844d7 Merge branch 'feature-signup'
da4bdb5 Added file6.html
5c4d7a0 (feature-signup) Added file2.html
5d764c8 Added file.html
8388cc5 Added python script ai_integration.py
9dbe926 (feature-login) Adding healthcheck script
a3ffa5c Added Login Page html file
484c603 Added code for login feature
539809b Update hello-world.sh
3ce969d Added index.html file
c3d31f2 Added script greet.sh
c4ba176 My First Shell Script
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git status
On branch master
nothing to commit, working tree clean
ubuntu@ip-172-31-30-155:~/devops-git-practice$
```

git reset --hard

commit is reverted

It wipes the changes from the staging area **and** deletes the physical edits in your files.

Reset Type	Moves HEAD?	Clears Staging Area?	Wipes Working Directory?
--soft	Yes	No	No
--mixed	Yes	Yes	No
--hard	Yes	Yes	Yes

Which one is destructive and why?

git reset --hard is the destructive one because it is the only one that modifies your **Working Directory**.

When would you use each one?

--soft

You committed, then realized you forgot to add one tiny file or had a typo in the commit message.

--mixed

You staged a bunch of files (**git add .**), but then realized you want to split them into three separate, cleaner commits.

--hard

if some security token is pushed

Should you ever use git reset on commits that are already pushed?

Because **git reset rewrites history**. If you reset and then "Force Push" (**git push --force**), you are deleting commits that your teammates might have already pulled onto their machines. This creates a "diverged history" nightmare for the rest of the team.

Task 2: Git Revert — Hands-On

Make 3 commits (commit X, Y, Z)

```
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git log --oneline
88646e9 (HEAD -> master) Modified login_page.html
54a6f57 Modified ai_integration.py
c9c5456 Modified login.html
0d38823 hotfix2
633b5cf added login.html
3a85323 Added file8.html
5a14da8 Added file7.html
d1e4082 Squashed commit of the following:
0d02246 Added file4.py
28844d7 Merge branch 'feature-signup'
da4bdb5 Added file6.html
5c4d7a0 (feature-signup) Added file2.html
5d764c8 Added file.html
8388cc5 Added python script ai_integration.py
9dbe926 (feature-login) Adding healthcheck script
a3ffa5c Added Login Page html file
484c603 Added code for login feature
539809b Update hello-world.sh
3ce969d Added index.html file
c3d31f2 Added script greet.sh
c4ba176 My First Shell Script
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git revert 54a6f57
[master 7a49ec7] Revert "Modified ai_integration.py"
1 file changed, 1 deletion(-)
```

Revert commit Y (the middle one) — what happens?

A new commit id is generated for the history that is reverted

```
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git revert 54a6f57
[master 7a49ec7] Revert "Modified ai_integration.py"
1 file changed, 1 deletion(-)
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git log --oneline
7a49ec7 (HEAD -> master) Revert "Modified ai_integration.py"
88646e9 Modified login_page.html
54a6f57 Modified ai_integration.py
c9c5456 Modified login.html
0d38823 hotfix2
633b5cf added login.html
3a85323 Added file8.html
5a14da8 Added file7.html
d1e4082 Squashed commit of the following:
0d02246 Added file4.py
28844d7 Merge branch 'feature-signup'
da4bdb5 Added file6.html
5c4d7a0 (feature-signup) Added file2.html
5d764c8 Added file.html
8388cc5 Added python script ai_integration.py
9dbe926 (feature-login) Adding healthcheck script
a3ffa5c Added Login Page html file
484c603 Added code for login feature
539809b Update hello-world.sh
3ce969d Added index.html file
c3d31f2 Added script greet.sh
c4ba176 My First Shell Script
ubuntu@ip-172-31-30-155:~/devops-git-practice$
```

Check git log — is commit Y still in the history?

Yes the commit is shown in the logs

Answer in your notes:

How is git revert different from git reset?

Git revert ,reverts the commit but adds a new commit id for the revert that is done.

Why is revert considered safer than reset for shared branches?

It doesnot rewrite history and the history is preserved.

When would you use revert vs reset?

If the commit is...	Use this command:	Why?
Local Only (Only on your laptop)	<code>git reset</code>	It keeps your history clean and linear before anyone else sees it.
Pushed/Public (On GitHub/Shared)	<code>git revert</code>	It prevents "merge conflicts" and history errors for your teammates.

Task 3: Reset vs Revert — Summary

Create a comparison in your notes:

	<code>git reset</code>	<code>git revert</code>
What it does	Can rewrite history.Moves the branch pointer to an earlier commit	Creates a new commit that undoes changes from a previous commit.Keeps original commit in history
Removes commit from history?	Yes	No
Safe for shared/pushed branches?	No	Yes
When to use	1.When you want to rewrite history or completely remove commits 2. On your local	1.On branches that are already pushed/shared.To undo a commit without breaking history 2. On Public repo/where you are contributing